

LAB 0

Aim: To study SQL commands their uses, syntax, and install MySQL for database management experiments

Q1)What is SQL and its Use?

Ans: SQL stands for **Structured Query Language**. It is a standard language used to communicate with and manage databases. SQL allows users to store, retrieve, modify, and manage data present in databases.

SQL was declared a standard by the **American National Standards Institute (ANSI)** in 1986 and by the **International Organization for Standardization (ISO)** in 1987.

Uses of SQL are as follows:

1. **Retrieving Data:** Used to fetch required information from one or more tables using commands such as SELECT.
2. **Adding Data:** New records can be inserted into tables using the INSERT command.
3. **Modifying Data:** Existing records can be changed with the help of the UPDATE command.
4. **Removing Data:** Records that are no longer required can be deleted using the DELETE command.
5. **Creating and Modifying Tables:** New tables can be created and existing tables can be changed using commands like CREATE TABLE and ALTER TABLE.
6. **Managing Database Objects:** SQL can also be used to manage schemas, indexes, and constraints for maintaining data integrity and improving database performance.

Q2) Describe all the different commands of SQL, mention use of it and syntax.

Ans: SQL commands are mainly divided into different categories according to their purpose.

1) DDL (Data Definition Language)

DDL commands are used to create, define, and modify the structure of database objects such as databases and tables.

a) CREATE DATABASE

Use: Creates a new database.

Syntax:

```
CREATE DATABASE database_name;
```

b) CREATE TABLE

Use: Creates a new table with specified columns and data types.

Syntax:

```
CREATE TABLE table_name (column1 datatype, column2 datatype, column3  
datatype);
```

c) ALTER TABLE

Use: Used to add, remove, or modify columns in an existing table.

Syntax:

```
ALTER TABLE table_name ADD column_name datatype;
```

d) DROP TABLE

Use: Permanently removes a table from the database.

Syntax:

```
DROP TABLE table_name;
```

e) TRUNCATE TABLE

Use: Removes all records from a table while keeping its structure.

Syntax:

```
TRUNCATE TABLE table_name;
```

f) CREATE INDEX

Use: Creates an index to make data searching faster.

Syntax:

```
CREATE INDEX index_name ON table_name(column_name);
```

g) DROP INDEX

Use: Deletes an existing index from the database.

Syntax:

```
DROP INDEX index_name;
```

2) DML (Data Manipulation Language)

DML commands are used to insert, update, and delete data stored in database tables.

a) INSERT INTO

Use: Adds new records to a table.

Syntax:

```
INSERT INTO table_name(column1, column2) VALUES(value1, value2);
```

b) UPDATE

Use: Changes existing records according to a specified condition.

Syntax:

```
UPDATE table_name SET column_name = value WHERE condition;
```

c) DELETE

Use: Removes selected records from a table.

Syntax:

```
DELETE FROM table_name WHERE condition;
```

3) DQL (Data Query Language)

DQL commands are mainly used for retrieving information from database tables.

a) SELECT

Use: Retrieves data from one or more tables.

Syntax:

```
SELECT column_name FROM table_name;
```

To retrieve all columns:

```
SELECT * FROM Student;
```

Common Clauses Used with SELECT:

WHERE

Use: Used to filter records according to a condition.

Syntax:

```
SELECT * FROM table_name WHERE condition;
```

ORDER BY

Use: Arranges the retrieved data in ascending or descending order.

Syntax:

```
SELECT * FROM table_name ORDER BY column_name ASC;
```

GROUP BY

Use: Combines rows having the same values into groups.

Syntax:

```
SELECT column_name, COUNT(*) FROM table_name GROUP BY column_name;
```

HAVING

Use: Applies a condition to grouped records.

Syntax:

```
SELECT column_name FROM table_name GROUP BY column_name HAVING  
condition;
```

4) DCL (Data Control Language)

DCL commands are used to manage permissions and access rights of database users.

a) GRANT

Use: Provides specific privileges to a user.

Syntax:

```
GRANT privilege ON table_name TO user_name;
```

b) REVOKE

Use: Takes back previously given privileges from a user.

Syntax:

```
REVOKE privilege ON table_name FROM user_name;
```

5) TCL (Transaction Control Language)

TCL commands are used to control database transactions and manage changes made to the database.

a) COMMIT

Use: Permanently saves the changes made during a transaction.

Syntax:

COMMIT;

b) ROLLBACK

Use: Cancels changes made after the last commit.

Syntax:

ROLLBACK;

c) SAVEPOINT

Use: Creates a temporary checkpoint within a transaction.

Syntax:

SAVEPOINT savepoint_name;

d) SET TRANSACTION

Use: Defines the properties or characteristics of a transaction.

Syntax:

SET TRANSACTION;

Q3) Write step by step procedure to install MYSQL.

Ans: **Step 1: Download MySQL Installer**

MySQL Community Downloads

MySQL Installer

MySQL Installer 8.0.40

Note: MySQL 8.0 is the final series with MySQL Installer. As of MySQL 8.1, use a MySQL product's MSI or Zip archive for installation. MySQL Server 8.1 and higher also bundle MySQL Configurator, a tool that helps configure MySQL Server.

Select Version:
8.0.40

Select Operating System:
Microsoft Windows

Platform	Version	Size	Download
Windows (x86, 32-bit), MSI Installer	8.0.40	2.1M	Download
Windows (x86, 32-bit), MSI Installer	8.0.40	306.5M	Download

We suggest that you use the MD5 checksums and GnuPG signatures to verify the integrity of the packages you download.

- Open a web browser on the computer.
- Visit the MySQL Community Downloads website.
- Download the MySQL Installer for Windows in .msi format.
- Select either the **Web Installer** or the **Full Installer** according to your requirement.

MySQL Community Downloads

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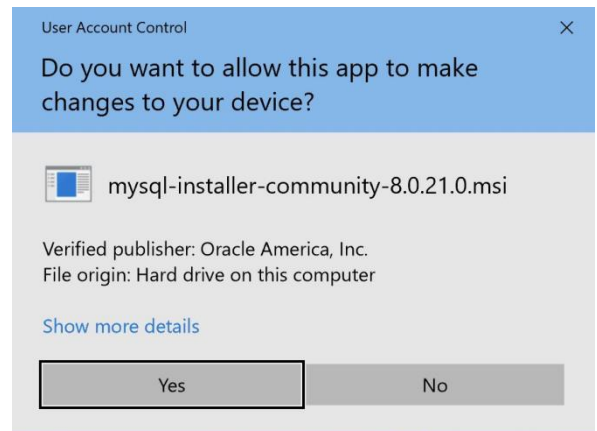
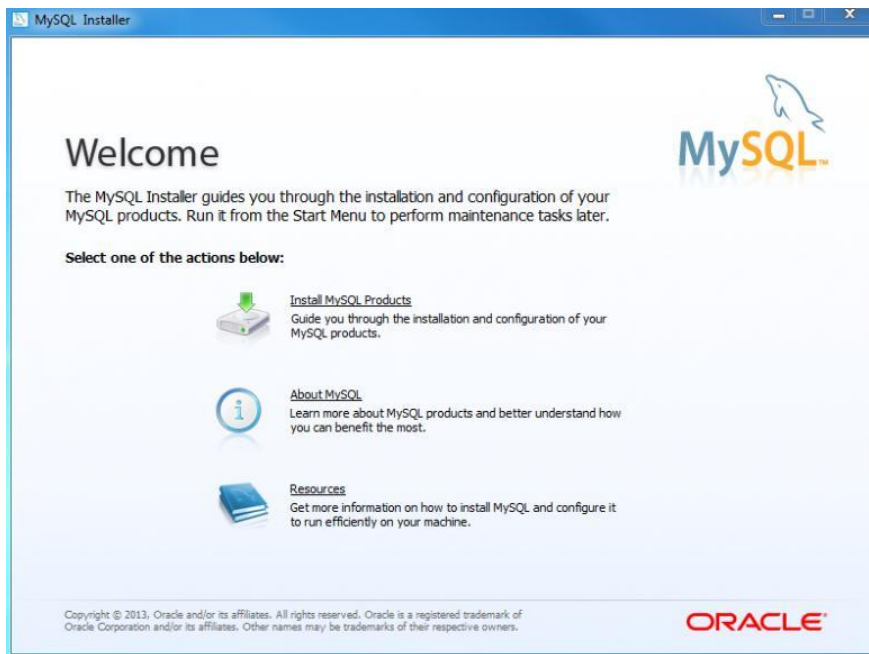
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for an Oracle Web account

MySQL.com is using Oracle SSO for authentication. If you already have an Oracle Web account, click the Login link. Otherwise, you can sign up for a free account by clicking the Sign Up link and following the instructions.

No thanks, just start my download.

Step 2: Start the Installer



- Locate and double-click the downloaded .msi file.
- If Windows asks for permission to make changes, select Yes.
- The MySQL Installer window will open.

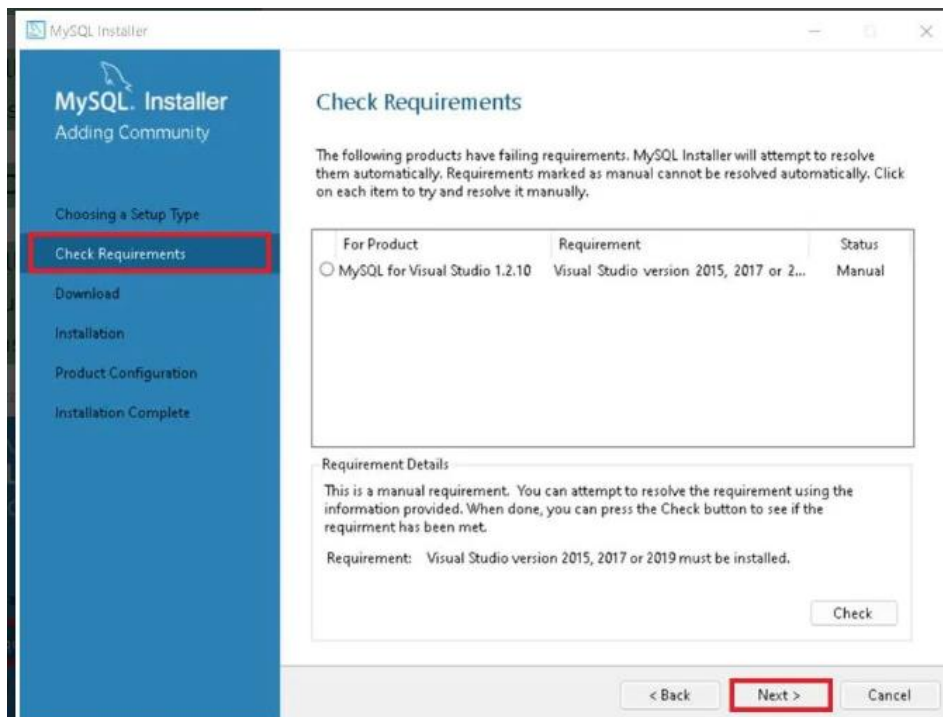
Step 3: Select Setup Type

- The installer will display different setup options.
- Select **Developer Default**, which is suitable for most users.

- Click **Next** to continue.

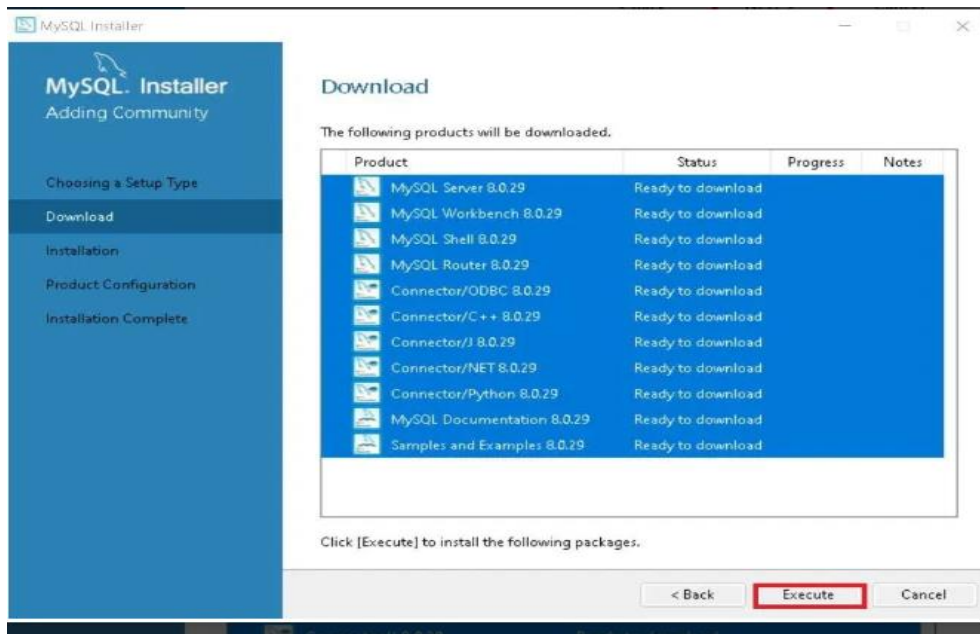
Step 4: Check Requirements

1. The installer will check for required dependencies such as the Microsoft Visual C++ Redistributable.
2. It may automatically resolve missing components, but in some cases, manual installation may be required.



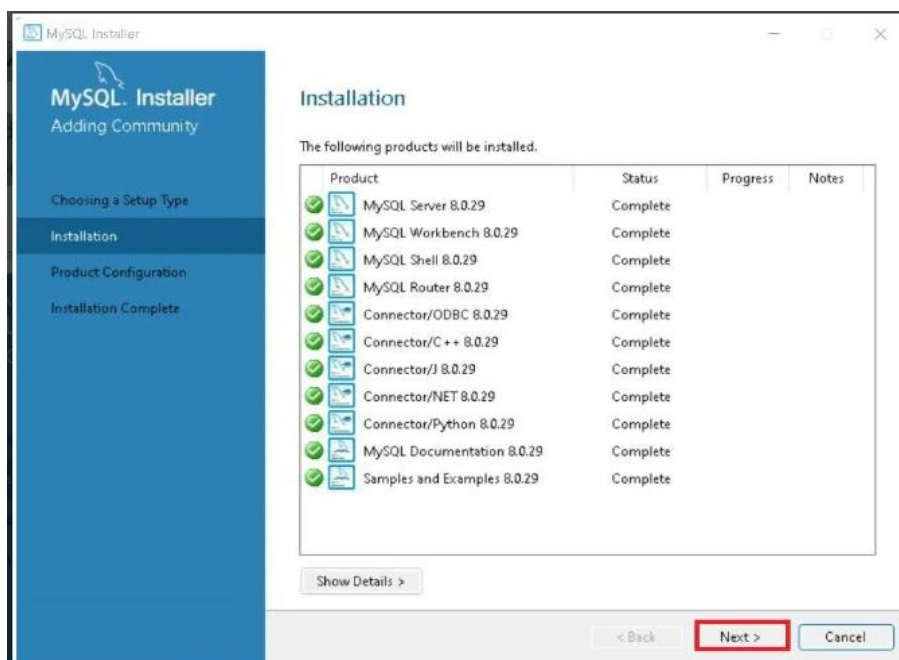
Step 5: Download MySQL Components

- In the download section, click **Execute** to begin downloading the selected components.
- Wait until the downloads are completed and the items display tick marks.
- Click **Next** to continue.



Step 6: MySQL Installation

- The downloaded components will now be installed on the system.
- Click **Execute** to begin the installation.
- Wait until the installation process is completed.
- Click **Next** to move ahead.

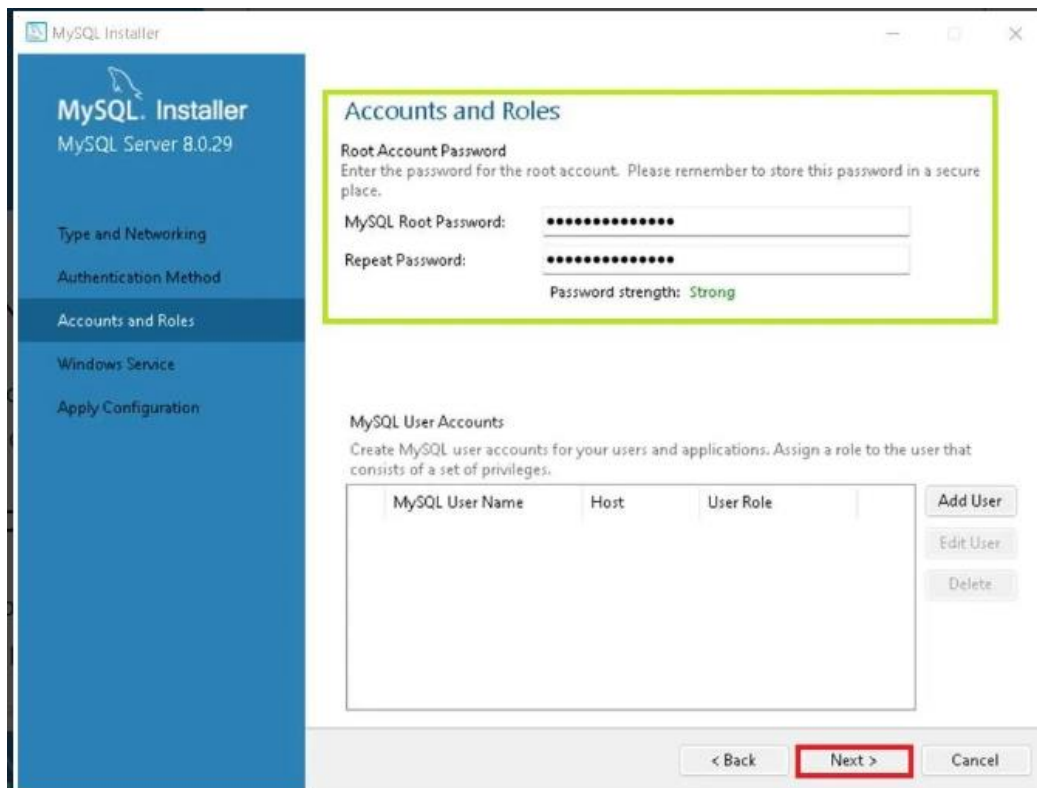


Step 7: Configure MySQL

- Continue through the configuration pages by clicking **Next**.
- Go through options such as **Type and Networking** and **Authentication Method**.
- Keep the required settings as instructed by the installer.

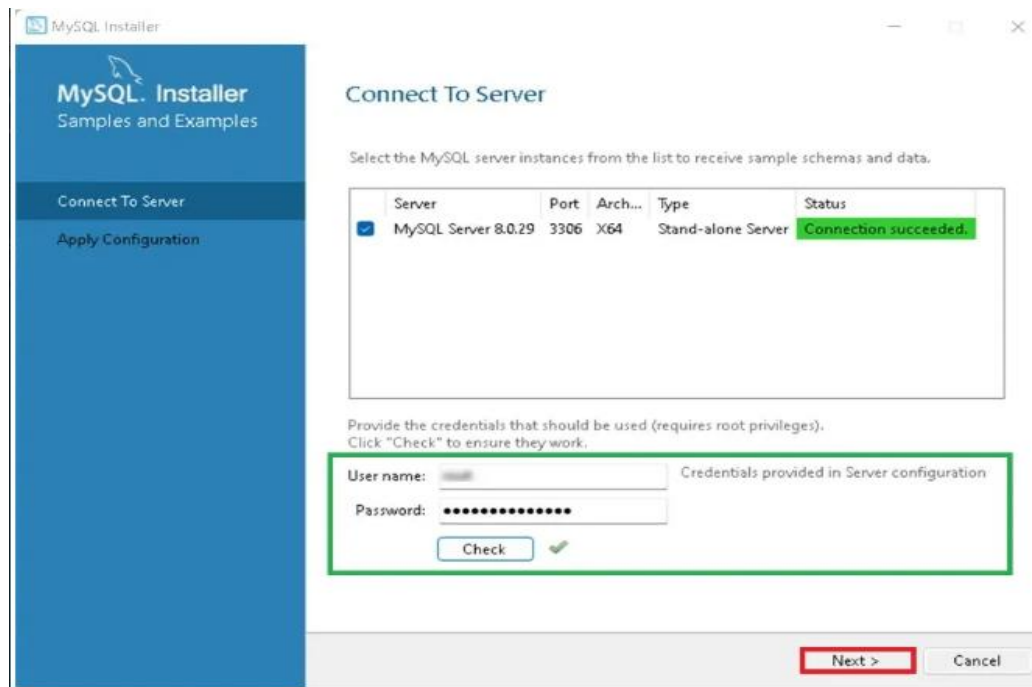
Step 8: Create MySQL Accounts

- Set a password for the **MySQL root user**.
- Choose a password that is strong and easy for you to remember.
- Click **Next** to continue.



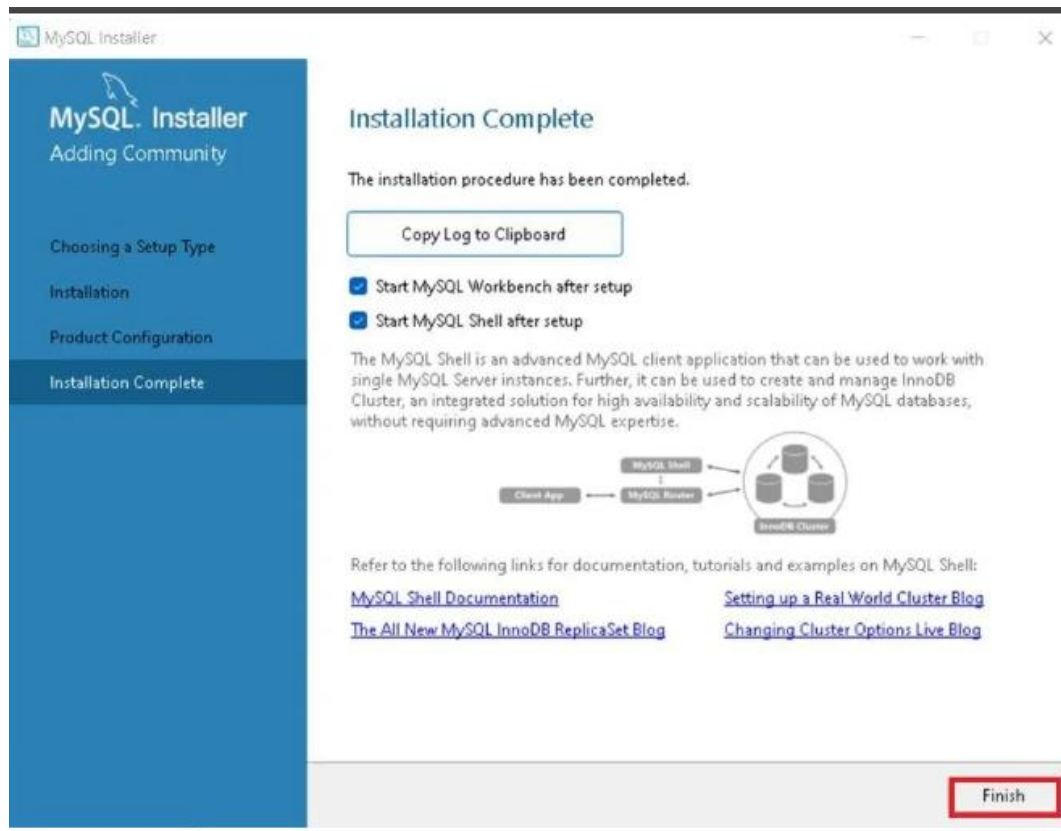
Step 9: Connect To Server

- Enter the root user's password.
- Click **Check** to test the connection.
- If the message shows “**Connection succeed**”, the server connection has been established successfully.



Step 10: Complete Installation

- After all installation and configuration steps are finished, click **Finish**.
- MySQL is now installed and configured on the Windows system.



Step 11: Verify MySQL Installation

1. To ensure a successful installation of MySQL, open the MySQL Command Line Client or MySQL Workbench, both available in your Start Menu.
2. Log in using the root user credentials you set during installation.